

1 Agroforestry SmartAgroforst: Poplar biomass modelling

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3 Abstract

Short-rotation agroforestry systems are increasingly promoted as a low-carbon feedstock option for bio-based value chains, but their economic viability depends on coordinated decisions on stand establishment, multi-cycle harvesting, and product cascading in spatially distributed supply chains. This paper develops a mixed-integer linear programming (MILP) model for the integrated design and operation of an agroforestry biomass supply chain with explicit age-lag constraints that link consecutive harvests at each site. The formulation couples allometric yield functions for different biomass types with a multi-product, multi-period network design model including storage, processing, and product quality cascades. The model is applied to a stylised case study of Central European poplar-based alley-cropping systems to explore trade-offs between harvest rotation length, product portfolios, and infrastructure configuration. We outline a computational experimentation plan to test the impact of allometric parameter uncertainty, price scenarios, and policy incentives on the optimal deployment of agroforestry systems in cleaner bioeconomy pathways.

4 *Keywords:* Agroforestry, Short-rotation coppice, Biomass supply chain, Product
5 cascading

6 1. Introduction

7 Agroforestry systems that integrate trees with agricultural crops (silvoarable systems)
8 are increasingly recognised as a promising land-use option to reconcile climate mitigation,
9 biodiversity conservation, and rural income generation on the same hectare of land (Toens-
10 meier, 2017). Temperate agroforestry systems (AFS) based on fast-growing species such

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11 as poplar, willow, alder, and black locust can deliver substantial biomass yields while im-
12 proving soil quality and providing multiple ecosystem services compared to conventional
13 monocultures (Huber et al., 2018). Biomasses produced in AFS is considered as a sus-
14 tainable feedstock and can be used either energetically, chemically or materially (e.g. in
15 the building sector). The latter two options are generally preferred over the energetic
16 usage from a cleaner production perspective as biogenic carbon is stored in products
17 during their life span which contributes to climate change mitigation.

18 Despite substantial research on ecological advantages and pioneering initiatives to
19 establish AFS in practice, the share of AFS in agricultural production systems is still
20 small. E.g. in Germany about 1200 hectares of silvoarable AFS are established in 2025,
21 which accounts for about 0.02% of total agricultural land use [DEFAP]. One of the core
22 challenges in establishing AFS systems at large scale in practice is a lack of opportunities
23 to generate substantial revenues from woody biomasses of AFS. On the other side, the
24 rise of bioeconomy creates substantial demands for woody biomass and requires a reliable
25 and cost-efficient feedstock supply. As woody biomasses are bulky feedstocks with low
26 energy density, the design of cost-efficient supply chains is crucial to ensure the economic
27 viability of AFS-based value chains. This requires coordinated decisions on stand estab-
28 lishment, multi-cycle harvesting, and product cascading in spatially distributed supply
29 chains, which can be supported by integrated optimisation models that link stand-level
30 growth dynamics to regional supply chain design.

31 In the last two decades, a large body of research has analysed biomass supply chains
32 for bioenergy and biorefineries using mathematical programming models, often in the
33 form of mixed-integer linear programming (MILP) (Zahraee et al., 2020; Sharma et al.,
34 2013). These models typically optimise network configuration, feedstock sourcing, and
35 logistics under cost or profit objectives. They often treat biomass availability as an exoge-
36 nous input or rely on simplified agricultural production decisions. Moreover, downstream
37 processing of biomasses is concentrated on one particular value chain e.g. bioenergy plants
38 consuming all produced biomasses. Recent work has started to integrate operational de-
39 tails such as various biomass types, biomass growth models and regeneration processes
40 in multi-period supply chain models.

41 The present paper contributes to this literature in four ways. First, we propose

a novel MILP formulation for AFS-based biomass supply chain design that explicitly links establishment decisions and consecutive harvest decisions, ensuring that harvesting decisions at each site are consistent with minimum and maximum rotation ages. Second, we embed allometric yield functions for different biomass types generating age-specific yields for different woody biomass types (like stems, branches, or leaves and twigs). Third, these different woody biomass types can be processed in specific downstream industries, like chemical or paper industry as well as energy sector. The model aims to provide strategic decision support for establishing regional AFS-based bioeconomy ecosystems connecting multiple industries and product markets.

The remainder of the paper is structured as follows. Section 2 reviews the relevant literature on agroforestry biomass systems, allometric biomass modelling, and biomass supply chain optimisation. Section 3 summarises the MILP formulation for the agroforestry supply chain with age-lag constraints. Section 4 outlines the planned computational experiments for the case study and discusses expected insights for cleaner production strategies.

2. Literature review

Agroforestry biomass systems, stand-level growth models, and biomass supply chain optimisation have each been studied extensively, but they are rarely combined in an integrated framework that links age-dependent stand dynamics to regional product-cascading value chains (Zahraee et al., 2020; Sharma et al., 2013; Espinoza et al., 2017). This section reviews (i) agroforestry systems with emphasis on system types, economics, and the cascading use of biomass as the primary revenue driver; (ii) biomass growth models at tree and stand level with explicit attention to the allometric–diameter–stand–biomass pathway; and (iii) biomass supply chain design using mathematical programming. The section concludes by identifying the modelling gap addressed in this paper.

2.1. Agroforestry systems

Agroforestry deliberately integrates woody perennials with crops and/or livestock on the same management unit, with the aim of enhancing productivity, profitability, and environmental performance (Nair, 1993; Toensmeier, 2017). In the temperate zone, three

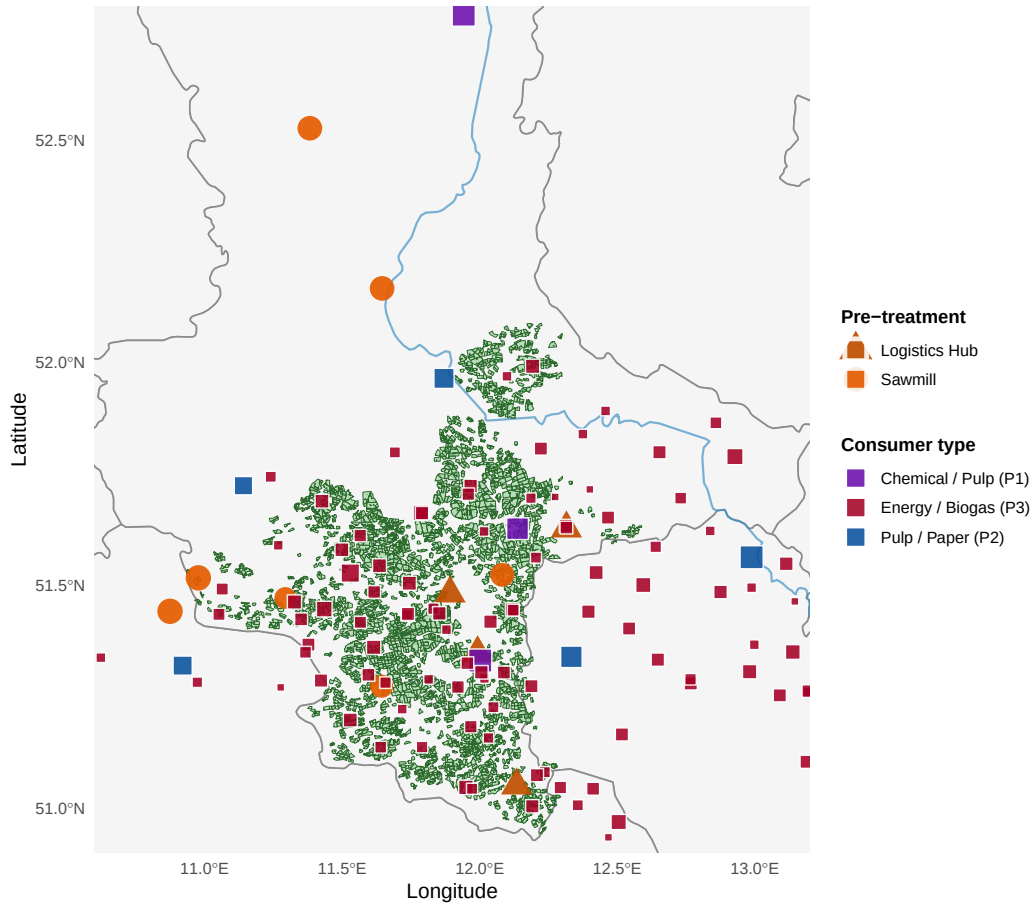


Figure 1: AFS supply chain network in Saxony-Anhalt. Grey lines: German federal state boundaries. Blue lines: major rivers. Green polygons: potential KUP sites. Orange symbols: pre-treatment facilities (circle = sawmill, triangle = logistics hub, size proportional to processing capacity). Coloured squares: consumer sites (violet: P1 chemical/pulp, blue: P2 pulp/paper, red: P3 energy/biogas, size proportional to total demand).

71 distinct system types can be distinguished based on the spatial arrangement of trees and
72 the rotation length at which woody biomass is harvested.

73 **Short-rotation coppice (SRC)** is defined as a monoculture plantation of fast-
74 growing woody species covering the entire management unit, with planting densities
75 of 8,000–20,000 cuttings $\cdot\text{ha}^{-1}$. After each harvest the root system regenerates without
76 replanting. Rotation cycles are typically 3–7~years and total stand lifetime is 20–25~years
77 (Trnka et al., 2008; Faasch and Patenaude, 2012). SRCs are no AFSs in the narrow sense,
78 but it is helpful to benchmark biomass production.

79 **Short-rotation agroforestry systems (SRAFS)** integrate SRC-type tree strips
80 (typically 1–4 rows of the same fast-growing species like poplar or willow) into an otherwise
81 arable field in an alley-cropping arrangement. Crop production continues in the inter-
82 row strips (Huber et al., 2018; Graves et al., 2007). Thus, tree strips typically cover
83 10–30% of the total parcel area. Rotation lengths and total stand lifetime in SRAFS are
84 comparable to SRCs. In SRAFS all trees are effectively edge trees, exposed to higher light
85 and water availability than interior SRC trees. This boundary effect raises individual-tree
86 biomass accumulation but also intensifies competition with adjacent crop rows (Huber
87 et al., 2018).

88 **Long-rotation agroforestry systems (LRAFS)** use the same alley-cropping spa-
89 tial structure as SRAFS but with rotation cycles of 12–15~years or longer (Graves et al.,
90 2007; Nair, 1993). At this rotation length, individual trees develop substantial stem di-
91 ameters and the proportion of high-value timber increases markedly, enabling premium
92 material uses (veneer, sawlogs, engineered wood products) that are inaccessible in short-
93 rotation systems. Therefore, often timber-quality tree species (such as walnut, cherry,
94 oak, or high-grade poplar clones) are planted.

95 SRAFS and LRAFS are not clearly differentiated in practice. First, a clear cut-off age
96 is hard to define and, mor importantly, the intended use of AFS biomass is not always
97 clear and can also change over time. Nonetheless, when establishing AFS, the structure
98 of AFS and its intended use after harvest drives planting decisions and, subsequently,
99 economic viability. Thus, in the present paper SRAFS and LRAFS are not considered as
100 separate AFS systems, but the interplay of intended use, establishment, and harvesting
101 decisions is analysed.

2.1.1. *Economic rationale of landowners*

The decision of a landowner to establish an AFS involves weighing substantial up-front costs and long-term opportunity costs against a diversified revenue stream. On the **cost side**, establishment requires site preparation, planting material, and planting labour, typically amounting to 1,500–3,500 €/ha for SRC or SRAFS (Faasch and Patenaude, 2012; Testa et al., 2014). Recurrent harvest and refitting costs (machine hire, chipping, transport to roadside) add approximately 200–400 €/ha per rotation depending on stand density and equipment availability (Testa et al., 2014; Spinelli et al., 2009). During the rotation, maintenance cost of AFS arise on a yearly basis. Next to expenses for establishment, maintenance and harvesting, opportunity cost arise as no revenues from crops are generated on tree strips. In regions with fertile soils and high cereal prices, yearly crop revenues may sum up to 400–600 €/ha (Faasch and Patenaude, 2012; Graves et al., 2007). Thus, opportunity cost are often the dominant cost driver in SRAFS profitability analyses.

On the **benefit side**, the primary revenue of AFS is the sale of harvested woody biomass, whose magnitude depends critically on rotation length, species, and site productivity. For poplar-based SRAFS in Central Europe, typical yields range from 10 to 25 t of dry matter biomass per ha for typical rotation lengths (Huber et al., 2018; Niemczyk and Thomas, 2021). Additionally, positive crop-yield effects in the inter-row strips of SRAFS and LRAFS may result. Tree rows act as windbreaks, reducing soil erosion and evapotranspiration; they provide partial shading that moderates heat stress during drought years; and their root systems may improve water retention in the topsoil (Graves et al., 2007; Grünewald et al., 2007). Empirical measurements in Central European alley-cropping trials indicate that these microclimate effects can increase cereal yields in adjacent strips by 5–15% relative to open-field controls, partially or fully compensating the loss of arable area occupied by the tree strips (Graves et al., 2007). Beyond direct yield effects, AFS deliver broader ecosystem services (like soil organic matter accumulation, carbon sequestration, biodiversity enhancement) that may lead to agri-environment subsidies and carbon credits (Toensmeier, 2017; Faasch and Patenaude, 2012).

Nonetheless, economic attractiveness of AFS depends primarily on revenues of AFS

133 biomass. Thereby, revenues are different for the types of biomass and in which value
 134 chain it is used. Often AFS biomasses are used energetically, i.e. for heat and power
 135 generation, or anaerobic digestion, which typically creates low prices per tonne of biomass.
 136 Alternatively, woody parts of the biomass (i.e., the stem and large branches) can also
 137 be used as pulp and paper grade material as well as feedstock for chemical industry
 138 or as timber achieving higher prices (Lamers et al., 2011; De Meyer et al., 2014). Thus,
 139 fostering AFS establishment by maximizing economic returns from AFS biomass requires
 140 a strategic approach to product cascading that allocates different biomass fractions to
 141 the highest-value applications, while ensuring that the rotation length is aligned with
 142 the growth dynamics of the stand and the market demand for different biomass types
 143 (Lamers et al., 2011).

144 Thus, for making educated decisions on AFS establishment and harvesting, the ex-
 145 pected biomass quantities at harvest for stem, branches, and residues need to be esti-
 146 mated. Thereby, the shares of the biomass types change with stand age. E.g.~the share
 147 of stem weight rises from roughly 55% at age 4 to over 75% at age 10 (Huber et al., 2018;
 148 Testa et al., 2014). Therefore, in Section 2.2 growth modelling approaches are reviewed
 149 that capture the age-dependent growth dynamics of different biomass types in AFS.

150 2.2. Biomass growth models

151 Total biomass on tree level is commonly modeled using allometric functions that
 152 relate tree characteristics like stem diameter or height to total biomass weight. There are
 153 numerous allometric models in the literature, often specific to species, site conditions, and
 154 stand age. For fast-growing species used in SRAFS (like poplar and willow), allometric
 155 functions of the power-law form are used [Reference]. I.e., dry biomass M_i (kg) of a shoot
 156 i is approximated by

$$M_i = b_0 \cdot SBD_i^{b_1} \quad (1)$$

157 where SBD is the shoot's stem base diameter (in cm) measured 10 cm above ground.
 158 Parameters b_0 and b_1 are species-specific (Huber et al., 2016). In the following it is
 159 assumed that each tree is pruned to one dominant shoot. For SRAFS species in Southern
 160 Germany, Huber et al. (2018) reports that most trees produce not more than 2 shoots
 161 and Huber et al. (2016) report a common exponent $b_1 = 2.603$ for black alder, black

locust, poplar (Max~3), and willow (Inger), while b_0 ranges from 0.025 to 0.041, reflecting differences in wood density and branching structure.

Allometric rules like (1) generate estimates of biomass quantities for a given shoot, i.e. they can be used to support operational harvest decisions. For planning on a strategic level, it is necessary to estimate biomass development over time, i.e. an SBD growth model. The mean stem base diameter $\overline{\text{SBD}}(t)$ of coppice shoots increases rapidly in the first growing seasons as shoots exploit stored root carbohydrates, then decelerates as intra-stand competition for light and water intensifies. Such trajectories can be described by sigmoidal growth functions. Empirical evidence from poplar SRC and SRAFS trials indicates that the SBD growth curve exhibits maximal growth rate (i.e., inflection point) at a relatively early stand age (2–4 years) (Barrio-Anta et al., 2008; Niemczyk and Thomas, 2021). Therefore, a Gompertz model for $\overline{\text{SBD}}(t)$ can be formulated as

$$\overline{\text{SBD}}(t) = D_\infty \cdot e^{-e^{-r \cdot (t-\tau)}}, \quad (2)$$

where D_∞ (cm) is the asymptotic mean diameter, r (yr^{-1}) controls the rate of diameter increment, and τ (yr) is the age at the inflection point of the diameter curve. Given $\overline{\text{SBD}}(t)$ from (2), the mean shoot biomass growth follows directly from the allometric equation (1):

$$\bar{M}_{\text{shoot}}(t) = b_0 \cdot \overline{\text{SBD}}(t)^{b_1} = d_\infty \cdot b_0 \cdot e^{-e^{-r \cdot b_1 \cdot (t-\tau)}}. \quad (3)$$

Thereby, $b_\infty = d_\infty \cdot b_0$ denotes the asymptotic mean shoot biomass, and the effective growth rate of shoot biomass is $b'_1 = r \cdot b_1$, which is amplified relative to the biomass growth rate r by the allometric exponent b_1 .

Total aboveground biomass quantities on stand level $M_{\text{AG}}(t)$ (in t per ha) is then obtained by aggregating individual shoot biomasses over all trees and multiplying by stand density $N(t)$ (in trees per ha):

$$M_{\text{AG}}(t) = N(t) \cdot \bar{n}_{\text{shoot}}(t) \cdot \bar{M}_{\text{shoot}}(t), \quad (4)$$

where $\bar{n}_{\text{shoot}}$ is the mean number of shoots per tree (Huber et al., 2018). Both $\bar{n}_{\text{shoot}}(t)$ and $N(t)$ may change over time. The number of shoots is often controlled in tree care processes and reduced to 1 or 2 shoots per tree (Huber et al., 2018). Stand level density $N(t)$ is initially defined when the AFS is established and often declines somewhat over

time, but usually saturates quickly [REFERENCE]. For the product $M_{AG}(t)$ in (4) a right-skewed sigmoidal shape with an inflection occurring at a relatively low fraction of the asymptote can be assumed which can be captured in a Gompertz model as well (Niemczyk and Thomas, 2021; Testa et al., 2014). Assuming constant tree density and an average shoot number of 1, $M_{AG}(t)$ can be simplified as

$$\bar{M}_{AG}(t) = a_{\infty} \cdot e^{-e^{-b'_1 \cdot (t-\tau)}}. \quad (5)$$

where $a_{\infty} = N \cdot \bar{n}_{shoot} \cdot d_{\infty} \cdot b_0$ is the asymptotic stand-level biomass. For poplar clone Max~3 under SRAFS conditions in Central Europe, representative parameter values are $D_{\infty} = 9.5$ cm, $r = 0.55$, and $\tau = 2.5$ years, yielding a mean SBD of approximately 4–5 cm at age 3 and 7–9 cm at age 7 which is consistent with the diameter ranges reported by Huber et al. (2018) and Barrio-Anta et al. (2008). For total aboveground biomass of poplar clone Max 3 under SRAFS management, the Gompertz model (5) is parameterised as $a_{\infty} = 100$ t of dry biomass per ha, $b'_1 = 0.32$ per year, and $\tau = 4.0$ years, yielding a growth peak at approximately age 4 and a total aboveground biomass of roughly 65 t of dry biomass per ha at age 10 which is consistent with the empirical reference range of 50–135 t per ha reported for comparable poplar systems in the literature (Huber et al., 2018; Niemczyk and Thomas, 2021).

Yet, for the purpose of modelling AFS supply chains, the three biomass components merchantable stem, structural branches, and fine residues need to be distinguished. Thus, the total biomass growth model (5) needs to be adjusted those types. Thereby, growth characteristics are different each type and, thus, require parameterization. Thereby, the differentiation between the merchantable stem and branch depends on the requirements defined by downstream processes. E.g. for chemical production, the merchantable stem is defined as the portion of the tree with a certain minimum diameter (e.g. 10 cm), while branches and residues are defined as the portion of the tree that cannot be chipped into such large chips and thus require different processing or are used for lower-value applications. The share of these components in total aboveground biomass changes with stand age, which has important implications for product cascading decisions in AFS-based supply chains.

Empirical studies show that the stem fraction rises from roughly 55% of AGB at age~4 to over 75% at age~10, while branch and residue shares decline as stands mature

(Huber et al., 2018; Testa et al., 2014, Jha (2018)). This age-dependence is biologically driven by the increasing dominance of secondary stem growth (radial increment) relative to primary shoot elongation and branching in later stand development stages. To capture these dynamics analytically, each component share $f_p(t)$ is represented as a normalised Gompertz function:

$$h_p(t) = a_p \cdot e^{-e^{-b_p \cdot (t - \tau_{0,p})}} \quad (6)$$

$$f_p(t) = \frac{h_p(t)}{\sum_{p'} h_{p'}(t)}. \quad (7)$$

For Max~3 the stem helper is parameterised with a high asymptote ($a_{\text{stem}} = 1.0$) and a relatively late inflection ($\tau_{0,\text{stem}} = 4.0$ years), whereas the branch uses a lower asymptote ($a_{\text{branch}} = 0.7$) and an earlier inflection ($\tau_{0,\text{branch}} = 3.0$ years), and the residue helper saturates earliest ($a_{\text{rest}} = 0.1$, $\tau_{0,\text{rest}} = 2.0$ years). This parameterisation reproduces the observed shift towards stem dominance with age and is consistent with the empirical component fractions compiled across poplar agroforestry and SRC systems spanning ages 6–13 years, as illustrated in Figure 2.

2.3. Biomass supply chain design

Biomass supply chain design problems have been extensively studied in literature (Zahraee et al., 2020; Sharma et al., 2013; Espinoza et al., 2017). Most models jointly optimise facility locations and capacities, biomass sourcing patterns, multimodal transport flows, and inventory policies under cost or profit objectives, sometimes augmented by greenhouse gas emission indicators (Ekşioğlu et al., 2009; Arabi et al., 2019). A key methodological feature of the more advanced formulations is the integration of long-term network design with multi-period logistics, enabling simultaneous optimisation of infrastructure and operational decisions across seasonal or annual time steps (Ekşioğlu et al., 2009; Zahraee et al., 2020).

Among generic multi-product frameworks, OPTIMASS explicitly models product quality cascades by defining a hierarchy of biomass classes and allowing higher-quality products to satisfy lower-quality demands, while the t-OPTIMASS extension incorporates biomass growth and regeneration over multiple periods (De Meyer et al., 2014, 2016). However, in virtually all existing formulations biomass availability is treated as

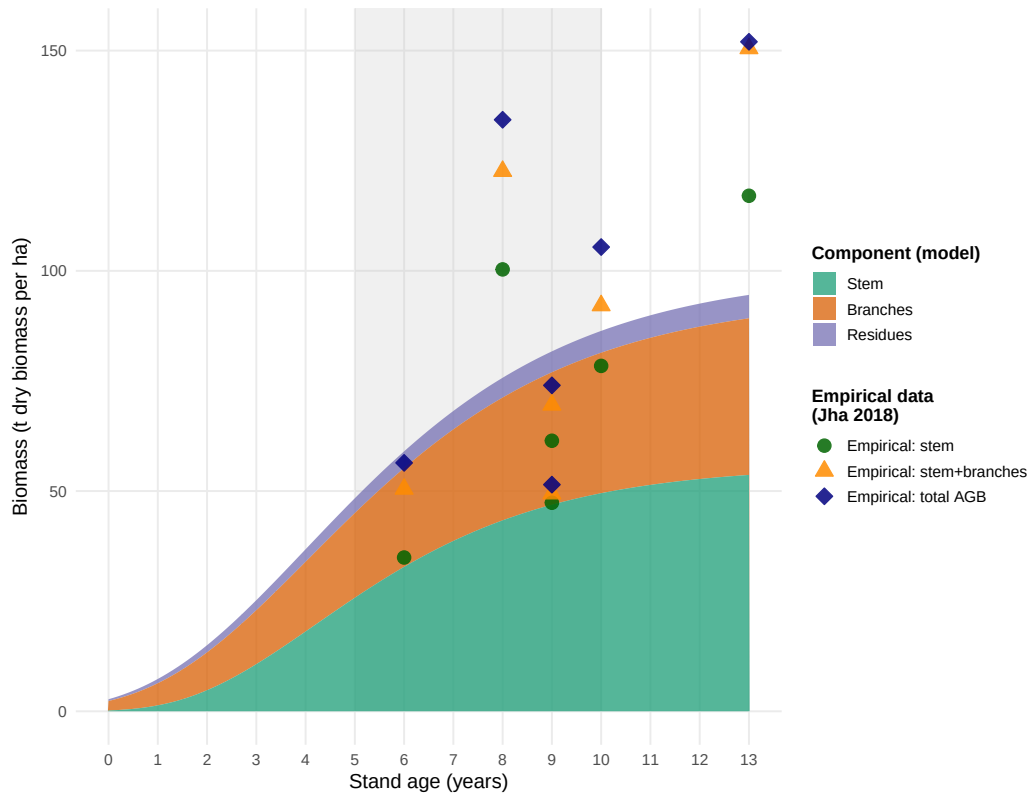


Figure 2: Modelled stand-level biomass components for poplar (Max 3) over stand age, represented as stacked Gompertz curves (shaded areas: stem green, branches orange, residues purple) and compared with empirical aboveground biomass observations from poplar agroforestry and SRC systems. Symbols denote cumulative component totals: stem (filled circles), stem+branches (triangles), total AGB (diamonds). Empirical data compiled from Jha (2018) and cited primary sources. Grey shading: typical rotation-age window (5–10 years).

an exogenous time-series input: stand age structure, rotation decisions, and coppice cycles are not represented endogenously, so that trade-offs between rotation length, stand management, and supply chain configuration cannot be explored in a unified model (De Meyer et al., 2016; Lamers et al., 2011). This gap is particularly pronounced for agroforestry systems with repeated coppice cycles, where establishment timing, consecutive harvest decisions, and age-dependent biomass quality jointly determine both the feasible supply profile and the achievable product cascade.

The MILP formulation developed in this paper addresses this gap by endogenising stand age dynamics through binary age-lag variables, coupling the component-specific yield coefficients η_{pa} derived from the Gompertz growth models above with supply chain flow variables, and embedding a product quality cascade that allocates stem, branch, and residue fractions to chemical, pulp, and energy markets within a single optimisation model.

2.4. Biomass supply chain design

The design and planning of biomass supply chains has attracted substantial research attention over the past two decades, driven by the need to ensure cost-efficient and sustainable feedstock provision for bioenergy and biorefinery applications. Several systematic reviews provide comprehensive overviews of this literature. Cambero and Sowlati (2014) survey economic, social, and environmental assessment and optimisation studies for forest biomass supply chains, identifying MILP as the dominant modelling approach and noting a growing trend towards multi-objective formulations integrating life cycle assessment.

Yue et al. (2014) review biomass-to-bioenergy and biofuel supply chain models with a focus on multi-scale modelling, uncertainty treatment, and sustainability metrics, and conclude that the coupling of upstream feedstock dynamics with downstream network design remains an underexplored area. Malladi and Sowlati (2018) survey logistics-specific features of biomass supply chain models—seasonal availability, moisture content degradation, preprocessing steps—and highlight that most models treat biomass as a homogeneous commodity, neglecting quality differentiation. Zahraee et al. (2020) provide a recent meta-review of 140+ studies, confirming that cost minimisation is still the dominant objective, and that multi-product cascading formulations are rare.

276 The structure of biomass supply chain models typically encompasses three decision
 277 layers: (i) strategic network design (facility location, capacity sizing, technology selec-
 278 tion), (ii) tactical planning (harvest scheduling, seasonal flows, inventory), and (iii) oper-
 279 ational routing. The following paragraphs discuss key individual contributions and their
 280 specific methodological advances.

281 **Strategic multi-product network design.** Ekşioğlu et al. (2009) develop one of
 282 the earliest large-scale MILP models for biomass-to-biorefinery supply chain design, inte-
 283 grating facility location, capacity sizing, and multimodal transport for lignocellulosic feed-
 284 stocks. Their model minimises total system cost and introduces binary facility-opening
 285 variables that have since become standard in the field. A key limitation, however, is that
 286 biomass supply is treated as an exogenous input, so feedstock availability is not endo-
 287 genised as a function of land-use or growth decisions. Mansoornejad et al. (2013) extend
 288 the strategic design perspective to forest biorefineries by integrating product portfolio
 289 design with supply chain configuration in a single MILP. Their model explicitly accounts
 290 for product conversion efficiencies along different processing pathways, enabling joint
 291 optimisation of process technology selection and logistics.

292 **Multi-period and tactical formulations.** De Meyer et al. (2014) introduce the
 293 OPTIMASS framework, a generic MILP for biomass-based supply chain optimisation
 294 that explicitly models product quality cascades: higher-quality biomass classes can sub-
 295 stitute for lower-quality demands downstream, whereas the reverse is infeasible. This
 296 cascade hierarchy, formalised through ordered product sets, is particularly relevant for
 297 woody biomass where stem, branch, and residue fractions command different market
 298 prices. The companion t-OPTIMASS extension (De Meyer et al., 2016) incorporates tem-
 299 poral dynamics by explicitly representing biomass growth and regeneration over multiple
 300 periods, enabling simultaneous optimisation of harvest timing and network flows. How-
 301 ever, in both OPTIMASS formulations the growth dynamics are represented as exogenous
 302 time-series rather than as age-structured stand models, so rotation-length decisions and
 303 age-dependent biomass quality are not endogenised. Arabi et al. (2019) demonstrate
 304 multi-period MILP extensions for bio-based supply chains with production dynamics,
 305 but focus on microalgae rather than woody biomass and do not address product quality
 306 cascades.

307 **Uncertainty and stochastic formulations.** Sharma et al. (2013) identify supply-
 308 side uncertainty (yield variability, seasonal availability, moisture content) as the most
 309 critical source of risk in biomass supply chains. In response, a strand of literature has
 310 developed stochastic and robust optimisation extensions. Zahraee et al. (2020) docu-
 311 ment that two-stage stochastic MILP and robust optimisation approaches now account
 312 for roughly 30% of the modelling literature, with biomass yield uncertainty and price
 313 volatility as the dominant stochastic parameters.

314 **Agroforestry and short-rotation coppice logistics.** Biomass supply chain re-
 315 search focusing specifically on agroforestry or SRC contexts is comparatively sparse. Es-
 316 pinoza et al. (2017) note that most supply chain models assume continuous feedstock
 317 availability and do not represent the periodic harvest cycles characteristic of coppice
 318 systems. Lamers et al. (2011) analyse feedstock design principles for lignocellulosic sup-
 319 ply chains and argue that the allocation of different biomass fractions to valorisation
 320 pathways—product cascading—is a key lever for economic viability, yet is rarely opti-
 321 mised jointly with network design decisions. The model developed in Section~3 of this
 322 paper directly addresses this gap by endogenising stand age dynamics through binary age-
 323 lag variables and coupling component-specific yield coefficients with a product quality
 324 cascade.

325 **Modelling features overview.** Table~1 summarises the key modelling features of
 326 the reviewed contributions.

327 **3. MILP model for agroforestry supply chain design**

328 *3.1. Sets, indices, and time structure*

329 The model considers a set of agroforestry sites $\mathcal{I}$, storage/processing facilities $\mathcal{J}$,
 330 consumer sites $\mathcal{K}$, and product types $\mathcal{P} = \{1, 2, 3\}$. Time is discretised into annual periods
 331 $t \in \mathcal{T} = \{1, \dots, T_{\max}\}$. To represent stand establishment and harvest timing, extended
 332 time sets $\mathcal{T}^+ = \{0, \dots, T_{\max} + 1\}$ and harvest periods $\mathcal{T}^{\text{harv}} = \{A^{\min} + 1, \dots, T_{\max}\}$ are
 333 defined, where $A^{\min}$ and $A^{\max}$ are minimum and maximum stand ages (years) at harvest.
 334 The set $\mathcal{S}$ comprises all ordered pairs (s, t) of consecutive harvest periods (including
 335 establishment at $s = 0$) satisfying the age-lag bounds:

$$\mathcal{S} = \{(s, t) \mid s, t \in \{0, \dots, T_{\max} + 1\}, t - s \in \{A^{\min}, \dots, A^{\max}\}\}. \quad (8)$$

Table 1: Key modelling features of selected biomass supply chain design studies.

Study	Objective	Multi-product	Quality cascade	Age dynamics	Uncertainty	Level
Eksioglu et al. (2009)	Cost min.	—	—	—	—	Strategic
Sharma & Taaffe (2013)	Cost min.	Partial	—	—	—	Strategic/Tactical
Mansoornejad et al. (2013)	Profit max.	✓	—	—	—	Strategic
Cambero & Sowlati (2014)	Cost/GHG/jobs	—	—	—	Stochastic	Strategic
Yue et al. (2014)	Cost/GHG	✓	—	—	—	Multi-scale
De Meyer et al. (2015)	Cost min.	✓	✓	Exogenous	—	Strategic/Tactical
De Meyer et al. (2016)	Cost min.	✓	✓	Exogenous TS	—	Multi-period
Lamers et al. (2015)	Cost/quality	✓	Partial	—	Scenario	Strategic
Espinoza et al. (2017)	Sustainability	✓	—	—	Scenario	Strategic
Malladi & Sowlati (2018)	Cost min.	Partial	—	—	—	Tactical/Oper.
Arabi et al. (2019)	Profit max.	✓	—	Exogenous	—	Multi-period
Zahraee et al. (2020)	Multi-obj.	✓	—	—	Stochastic	Strategic/Tactical
This paper	NPV max.	✓	✓	Endogenous	Scenario	Strategic/Multi-period

3.2. Decision variables

Binary variables z_{ist} indicate whether, at site i , periods s and t form consecutive harvests (or establishment and first harvest if $s = 0$), with $(s, t) \in \mathcal{S}$ as defined in (8). These variables define a path of consecutive harvest cycles at each site over the planning horizon. Continuous variables $Y_{ipt} \geq 0$ represent the maximum harvest quantity (t) of product p at site i in harvest period t . Flow variables X_{ijpt} and $X_{jkpp't}$ represent transport of products from sites to storage and from storage to consumers, respectively, and S_{jpt} represent inventories at storage locations.

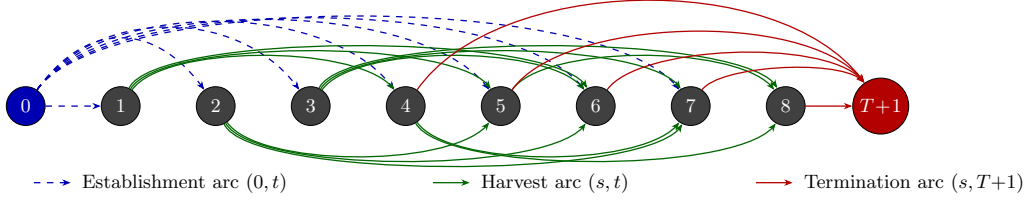


Figure 3: Network structure of set $\mathcal{S}$ for $T = 8$, $A_{\min} = 3$, $A_{\max} = 5$. Dashed blue arcs: establishment arcs $(0, t)$, $t \in \{1, \dots, T-1\}$. Solid green arcs above: harvest arcs (s, t) with odd s ; below: harvest arcs with even s . Solid red arcs: termination arcs $(s, T+1)$, $s \in \{A_{\min} + 1, \dots, T\}$.

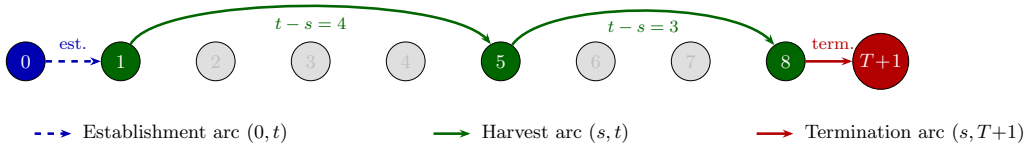


Figure 4: Example of a feasible path through the network for $T = 8$, $A_{\min} = 3$, $A_{\max} = 5$: establishment in period 1, harvests in periods 5 and 8, termination at $T + 1$.

3.3. Objective function

The model maximises net present value over the planning horizon, expressed as total revenues minus establishment, opportunity, harvest, transport, and storage costs:

$$\begin{aligned}
 \max Z = & \sum_{k,p,t} R_{k,p} \cdot \sum_{j \in \mathcal{J}, p' \in \mathcal{P}: p' \leq p} X_{jkp'pt} \\
 & - \sum_i C_i^{\text{est}} \cdot \text{AREA}_i \cdot \sum_{t \in \mathcal{T}} z_{i0t} \\
 & - \sum_i C_i^{\text{opp}} \cdot \text{AREA}_i \cdot \sum_{t \in \mathcal{T}} (T_{\max} - t) \cdot z_{i0t} \\
 & - \sum_{i,(s,t) \in \mathcal{S}: s > 0} C_i^{\text{harv}} \cdot \text{AREA}_i \cdot z_{ist} \\
 & - c^{\text{tr-raw}} \sum_{i,j,p,t} d_{ij} \cdot X_{ijpt} - c^{\text{tr-pre}} \sum_{j,k,p,t} d_{jk} \cdot X_{jkpt} - \sum_{j,p,t} c_j^{\text{stor}} \cdot S_{jpt}.
 \end{aligned} \tag{9}$$

Here, $R_{k,p}$ are product- and consumer-specific revenues (€/t), C_i^{est} are establishment costs per hectare, C_i^{opp} are annual opportunity costs per hectare, and C_i^{harv} are harvest and refitting costs per hectare (Testa et al., 2014). Transport costs are proportional to flow and distance with rates $c^{\text{tr-raw}}$ and $c^{\text{tr-pre}}$ for raw and pre-treated biomass, respectively, and storage incurs linear holding costs c_j^{stor} per tonne.

3.4. Key constraints

The path connectivity constraint ensures that each harvest period at a site has exactly one predecessor and one successor in the age-lag graph, effectively forming a path over $\mathcal{T}^+$:

$$\sum_{s=0:(s,t) \in \mathcal{S}}^t z_{ist} = \sum_{u=t+1:(t,u) \in \mathcal{S}}^{T_{\max}+1} z_{itu} \quad \forall i \in \mathcal{I}, t \in \mathcal{T}. \quad (10)$$

Establishment is restricted to at most one time per site:

$$\sum_{t \in \mathcal{T}^+} z_{i0t} \leq 1 \quad \forall i \in \mathcal{I}. \quad (11)$$

Biomass yield is linked to age-lag decisions through the allometric yield coefficients:

$$Y_{ipt} = \sum_{s=1}^{t-1} \eta_{p(t-s)} \cdot \text{AREA}_i \cdot z_{ist} \quad \forall i \in \mathcal{I}, p \in \mathcal{P}, t \in \mathcal{T}^{\text{harv}}. \quad (12)$$

Harvest quantities bound outbound flows from each site:

$$\sum_{j \in \mathcal{J}} X_{ijpt} \leq Y_{ipt} \quad \forall i, p, t \in \mathcal{T}^{\text{harv}}. \quad (13)$$

At storage hubs, inventories obey standard balance equations with initial inventory zero:

$$S_{jpt} = S_{jp,t-1} + \sum_i X_{ijpt} - \sum_{k,p' \geq p} X_{jkpp't} \quad \forall j, p, t \in \mathcal{T}^{\text{harv}}. \quad (14)$$

Storage and processing capacities are enforced via:

$$\sum_p S_{jpt} \leq \text{CAP}_j^{\text{stor}} \quad \forall j, t, \quad \sum_{i,p} X_{ijpt} \leq \text{CAP}_j^{\text{proc}} \quad \forall j, t. \quad (15)$$

Demand satisfaction with cascades is modelled as described above, where higher-quality products can satisfy lower-quality demands up to maximum demand levels $D_{kpt}^{\max}$.

4. Case study and computational experiment plan

The proposed MILP model is intended to support the design of agroforestry biomass supply chains for cleaner production in specific regional contexts. In a forthcoming case study, we focus on poplar-based alley-cropping systems in a temperate European setting, informed by empirical data from SRAFS trials in Southern Germany and SRC plantations in Central Europe (Huber et al., 2018; Trnka et al., 2008; Testa et al., 2014).

370 We envisage the following steps and experimental scenarios for computational analy-
371 sis:

- 372 **1. Parameterisation of allometric yield profiles:** Use species- and site-specific
373 allometric models and observed growth trajectories to construct age-dependent
374 yield coefficients η_{pa} for chemical, pulp, and energy product classes at representative
375 agroforestry sites. Where necessary, scenario ranges will be defined to capture
376 uncertainty in MAI and CAI due to climate variability and management differences
377 (Huber et al., 2016, 2018; Trnka et al., 2008).
- 378 **2. Baseline optimisation:** Solve the MILP model (9)–(15) for a baseline configu-
379 ration with fixed prices, capacities, and policy settings, obtaining optimal estab-
380 lishment timing, harvest schedules, and supply chain structure (locations, flows,
381 inventories).
- 382 **3. Rotation length scenarios:** Vary the age-lag bounds $A^{\min}$ and $A^{\max}$ in (8) to
383 represent alternative rotation strategies (e.g., 3–5 years vs. 8–10 years) and analyse
384 their impacts on profitability, product mix, and land use. This will shed light on
385 trade-offs between shorter rotations favouring energy chips and longer rotations
386 enabling higher shares of material products (Trnka et al., 2008; Testa et al., 2014).
- 387 **4. Product price and cascading scenarios:** Explore scenarios with different rel-
388 ative prices for chemical, pulp, and energy products, reflecting changes in market
389 demand and policy incentives (e.g., bioenergy subsidies, green chemicals markets).
390 Assess how the product cascade hierarchy and yield quality structure influence the
391 marginal value of species and rotation decisions (De Meyer et al., 2014; Lamers
392 et al., 2011).
- 393 **5. Infrastructure and logistics scenarios:** Evaluate the effect of alternative stor-
394 age and processing network configurations (e.g., decentralised vs. centralised hubs,
395 different capacity levels, transport cost parameters) on the optimal spatial deploy-
396 ment of agroforestry sites and biomass flows (De Meyer et al., 2014, 2016; Lamers
397 et al., 2011).
- 398 **6. Sensitivity to allometric and growth uncertainty:** Conduct parametric sen-
399 sitivity and, where computationally feasible, stochastic analyses to quantify the
400 impact of uncertainty in allometric parameters and growth responses (e.g., under

drought or management differences) on optimal decisions and system robustness (Huber et al., 2018; Trnka et al., 2008).

7. Cleaner production indicators: For selected solutions, compute derived indicators such as biomass cascading index (share of biomass used for material products before energy), utilisation rates of higher-quality fractions, and, where coupled with LCA data, greenhouse gas emission savings relative to fossil benchmarks (Espinoza et al., 2017; Zahraee et al., 2020).

5. Conclusions

This paper outlines a research framework that couples allometric biomass growth models (1)–(7) with a MILP-based agroforestry biomass supply chain design model (9)–(15) featuring age-lag constraints and product cascading. Building on empirical evidence from temperate SRAFS and SRC systems and established biomass supply chain models, the proposed approach aims to capture key interactions between stand-level growth dynamics and network-level design decisions that are central to the cleaner production potential of agroforestry (Huber et al., 2018; De Meyer et al., 2014, 2016). Future work will implement and calibrate the model for a regional case study, conduct the computational experiments sketched above, and extend the framework towards multi-objective optimisation including environmental indicators such as greenhouse gas emissions and biodiversity proxies.

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